Attendance Recovery Computation

Problem statement

Calculate current attendance and find the minimum consecutive future classes a student must attend to reach an eligibility threshold under different future attendance scenarios.

Assumptions and rules

• Attendance percentage = classes attended / classes held × 100.

• The student attends every recovery class in the main calculation.

• A second scenario allows a fixed number of future absences; the required attended classes are calculated first, then absences are added to the total held count.

• Eligibility is reached when the percentage is greater than or equal to the threshold.

Algorithm

1. Read classes held, classes attended, and eligibility percentage.

2. Validate that attended classes are between zero and classes held.

3. Calculate current attendance percentage.

4. If current percentage already meets the threshold, required classes are zero.

5. Otherwise test future attended-class counts from 1 upward until the projected percentage meets the threshold.

6. Repeat the calculation for a scenario with planned future absences.

7. Display current percentage and minimum recovery classes.

Pseudocode

```
START\nINPUT held, attended, threshold\nVALIDATE 0 <= attended <= held\ncurrent = attended / held * 100\nIF current >= thres
```

Flowchart

```
[Start] -> [Read held, attended, threshold]\n   |\n[Validate input and calculate current %]\n   |\n{Already eligible?} --Ye
```

Python solution

The program uses beginner-level Python constructs from the assigned syllabus: variables, data types, operators, conditions, loops, strings, lists, tuples, and unpacking where appropriate.

```
classes_held = 80
classes_attended = 54
threshold = 75
planned_absences = [0, 2, 5]

if classes_held <= 0:
    print("Invalid number of classes held")
elif classes_attended < 0 or classes_attended > classes_held:
    print("Invalid attendance data")
elif threshold <= 0 or threshold > 100:
    print("Invalid threshold")
else:
    current = classes_attended / classes_held * 100
    print("Current attendance:", round(current, 2), "%")

    for absences in planned_absences:
        future_attended = 0
        while True:
            projected = ((classes_attended + future_attended) /
                         (classes_held + future_attended + absences)) * 100
            if projected >= threshold:
                break
            future_attended = future_attended + 1
```

```
        print("If future absences =", absences,
              "minimum classes to attend =", future_attended,
              "projected attendance =", round(projected, 2), "%")
```

Sample output

```
Current attendance: 67.5 %\nIf future absences = 0 minimum classes to attend = 24 projected attendance = 75.0 %\nIf future a
```

Explanation / observations

The loop searches upward from zero and stops at the first qualifying value, so the answer is the minimum number of future classes to attend. Planned absences are assumed to occur in addition to the recovery classes.

Conclusion

This solution models the stated case using a clear sequence of input, processing, decisions, repetition, and output. The assumptions and sample data can be changed to test other cases.